Assignments 02

1.Significant earthquakes since 2150 B.C.

The Significant Earthquake Database contains information on destructive earthquakes from 2150 B.C. to the present. On the top left corner, select all columns and download the entire significant earthquake data file in .tsv format by clicking the Download TSV File button. Click the variable name for more information. Read the file (e.g., earthquakes-2026-10-14_11-23-28_+0800.tsv) as an object and name it Sig_Eqs.

1.1 [5 points] Compute the total number of deaths caused by earthquakes since 2150 B.C. in each country, and then print the top 20 countries along with the total number of deaths.

In [33]:
```
from turtle import down
import pandas as pd
#导入数据集
Sig_Eqs = pd.read_csv(
    "earthquakes-2026-10-14_21-39-07_+0800.tsv",
    sep="\t",
    skiprows=[1]
)

#计算总死亡数不包括次生灾害
top20_deaths = (
    Sig_Eqs.groupby("Country")["Deaths"] #分组
    .sum()#求和
    .sort_values(ascending=False)#降序排序
    .head(20)#选择前20个
)

print(top20_deaths) #打印
```

```
Country
CHINA           2139209.0
TURKEY          1199720.0
IRAN            1014453.0
ITALY            492236.0
SYRIA            419826.0
HAITI            323484.0
AZERBAIJAN       319251.0
JAPAN            242516.0
ARMENIA          177890.0
PAKISTAN         145085.0
IRAQ             136200.0
ECUADOR          135496.0
TURKMENISTAN     117412.0
PERU             102309.0
ISRAEL            93388.0
PORTUGAL          83572.0
GREECE            81162.0
CHILE             64277.0
INDIA             63507.0
TAIWAN            57154.0
Name: Deaths, dtype: float64
```

1.2 [10 points] Compute the total number of earthquakes with magnitude larger than 3.0 (use column Ms as the magnitude) worldwide each year, and then plot the time series. Do you observe any trend? Explain why or why not?

In [ ]:

```python
from turtle import down
import pandas as pd
#导入数据集
Sig_Eqs = pd.read_csv(
    "earthquakes-2026-10-14_21-39-07_+0800.tsv",
    sep="\t",
    skiprows=[1]
)

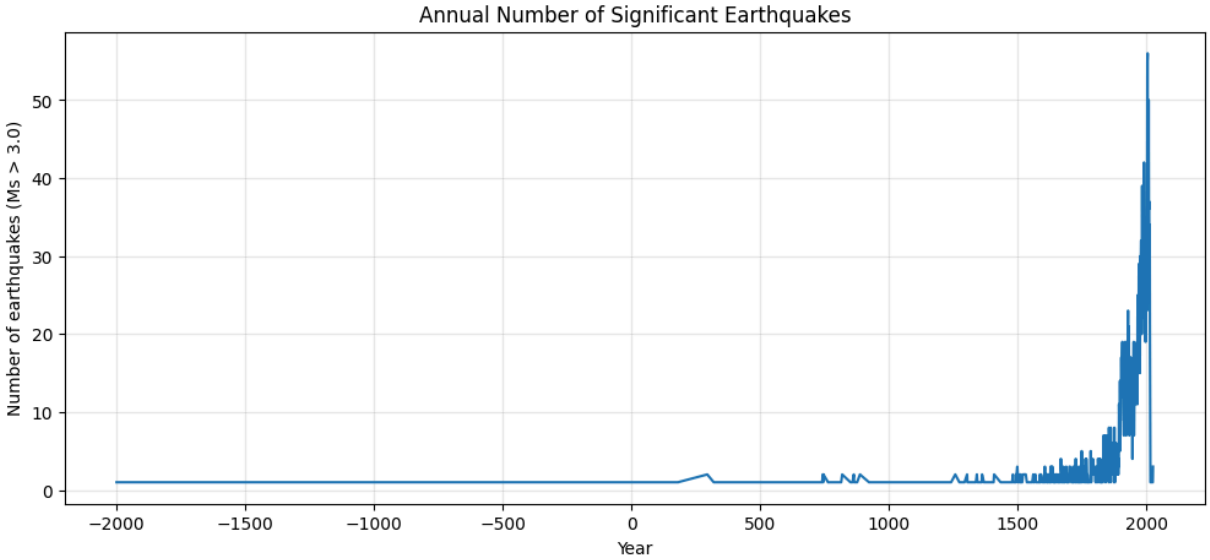
#计算记录大于3的数量
earthquakes_per_year  = (
    Sig_Eqs[Sig_Eqs["Ms"]>3.0]#筛选大于3的地震
    .groupby("Year") #分组
    .size()#求记录总数
)

import matplotlib.pyplot as plt

ax = earthquakes_per_year.plot(figsize=(12, 5))

ax.set_xlabel("Year")#x轴标题
ax.set_ylabel("Number of earthquakes (Ms > 3.0)")#y轴标题
ax.set_title("Annual Number of Significant Earthquakes")#标题
ax.grid(alpha=0.3)

plt.show()
```



从图中可以看到在公元1500前地震数量较少，在这之后呈现增加趋势，特别是20世纪以后增加明显，同时也存在明显波动。最后面突然降低是因为没有记录到完整年份。 但是，这不能直接证明真实地震发生频率一直在增加。古代地震主要依赖历史资料，记录不完整；近现代地震仪器、监测台站和资料收集系统不断完善，使更多地震能够被探测和记录。此外，Ms 震级数据本身的可用性也会影响统计结果。因此，图中趋势很大程度上可能反映观测和记录能力的变化，而不完全是地震活动本身的变化。

1.3 [10 points] Write a function CountEq_LargestEq that returns (a) the total number of earthquakes since 2150 B.C. in a given country AND (b) date and location of the largest earthquake ever happened in this country. Apply CountEq_LargestEq to every country in the file, report your results in a descending order.

In [54]:
```
import pandas as pd

Sig_Eqs = pd.read_csv(
    "earthquakes-2026-10-14_21-39-07_+0800.tsv",
    sep="\t",
    skiprows=[1]
)  # 读取 TSV 文件，跳过第二行搜索参数


def CountEq_LargestEq(Con):
    country_data = Sig_Eqs[Sig_Eqs["Country"] == Con]  # 筛选指定国家
    num = len(country_data)  # 统计地震记录总数

    if country_data["Ms"].notna().any():  # 检查该国是否有 Ms 记录
        largest_eq = country_data.loc[country_data["Ms"].idxmax()]  # 取 Ms 最大的记
        largest_Y = int(largest_eq["Year"])  # 提取年份并转为整数
        largest_M = int(largest_eq["Mo"]) if pd.notna(largest_eq["Mo"]) else "?"  #
        largest_D = int(largest_eq["Dy"]) if pd.notna(largest_eq["Dy"]) else "?"  #
        largest_date = f"{largest_Y}-{largest_M}-{largest_D}"  # 合并日期
        largest_pos = largest_eq["Location Name"]  # 提取地震位置
    else:
        largest_date, largest_pos = "Unknown", "Unknown"  # 没有 Ms 时无法确定最大地
```

```
    return Con, num, largest_date, largest_pos  # 返回该国的统计结果


results = []  # 创建列表保存各国结果
for country in Sig_Eqs["Country"].dropna().unique():  # 遍历所有不重复国家
    results.append(CountEq_LargestEq(country))  # 调用函数并保存结果

all_countries = pd.DataFrame(  # 将结果转换为 DataFrame
    results,
    columns=["Country", "Total Earthquakes", "Largest Date", "Largest Location"]
)

all_countries = all_countries.sort_values(  # 按地震总数降序排列
    by="Total Earthquakes",
    ascending=False
).reset_index(drop=True)  # 重置行索引

print(all_countries.to_string(index=False))  # 输出所有国家的结果
```

Country	Total Earthquakes	Largest Date	Largest Location
CHINA	626	1920-12-16	CHINA: GANSU PROVINCE, SHANXI PROVINCE
JAPAN	429	869-7-13	JAPAN: SANRIKU
INDONESIA	427	2004-12-26	INDONESIA: SUMATRA: ACEH: OFF WEST COAST
IRAN	385	856-12-22	IRAN: DAMGHAN, QUMIS
TURKEY	359	1939-12-26	TURKEY: ERZINCAN
ITALY	338	1915-1-13	ITALY: MARSICA, AVEZZANO, ABRUZZI
GREECE	296	365-7-21	GREECE: CRETE: KNOSSOS
USA	283	1957-3-9	ALASKA
PHILIPPINES	233	1897-9-21	PHILIPPINES: MINDANAO, ZAMBOANGA, SULU, ISABELA
MEXICO	217	1787-3-28	MEXICO: SAN MARCOS, OAXACA
CHILE	200	1730-7-8	CHILE: VALPARAISO
PERU	195	1619-2-14	PERU: TRUJILLO, PIURA, SANTA
RUSSIA	159	1952-11-4	RUSSIA: KAMCHATKA PENINSULA
PAPUA NEW GUINEA	107	2000-11-16	PAPUA NEW GUINEA: NEW IRELAND, DUKE OF YORK
TAIWAN	103	1920-6-5	TAIWAN
INDIA	102	1897-6-12	INDIA: ASSAM; BANGLADESH
COLOMBIA	83	1904-1-20	PANAMA-COLOMBIA
AFGHANISTAN	72	1909-7-7	AFGHANISTAN: HINDU-KUSH
NEW ZEALAND	72	1826-?-?	NEW ZEALAND: FJORDLAND
ECUADOR	70	1906-1-31	ECUADOR: OFF COAST
VENEZUELA	65	1900-10-29	VENEZUELA: MACUTO
VANUATU	65	1910-6-16	VANUATU ISLANDS
SOLOMON ISLANDS	63	1934-7-18	SOLOMON ISLANDS: SANTA CRUZ ISLANDS
ALGERIA	58	1980-10-10	ALGERIA: NORTHERN
ALBANIA	56	1893-6-14	ALBANIA: HIMARA, DHERMI, KUC, KUDHESI, VLORE, KANINA
PAKISTAN	55	1945-11-27	PAKISTAN: MAKRAN COAST
CROATIA	53	1667-4-6	BALKANS NW: CROATIA: DUBROVNIK: RAGUSA

	GUATEMALA	47	1942-8-6	
GUATEMALA: NEAR S COAST				
	MYANMAR (BURMA)	43	1912-5-23	
MYANMAR (BURMA): MANDALAY, MOGOK, MAYMYO				
	FRANCE	43	1909-6-11	FRA
NCE: VERNEGUES,CHARLEVAL,LA ROQUE D'ANTHERON				
	EL SALVADOR	42	1915-9-7	
EL SALVADOR				
	NICARAGUA	40	1898-4-29	
NICARAGUA: LEON, CHINANDEGA, MANAGUA				
	ARGENTINA	39	1894-10-27	
ARGENTINA: LA RIOJA, SAN JUAN, MENDOZA				
	USA TERRITORY	39	1902-9-22	
GUAM: AGANA				
	COSTA RICA	37	1950-10-5	
COSTA RICA-NICARAGUA				
	SPAIN	35	1884-12-25	
SPAIN: ARENAS DEL REY, VEGA, ALHAMA, MALAGA				
	SYRIA	34	1202-5-20	
SYRIA: SOUTHWESTERN				
	AUSTRALIA	34	1989-5-23	MACQ
UARIE ISLAND: MACQUARIE STATION; NEW ZEALAND				
	SWITZERLAND	31	Unknown	
Unknown				
	TAJIKISTAN	29	1907-10-21	
TAJIKISTAN: KARATAG				
	PORTUGAL	29	1761-3-30	
PORTUGAL: LISBON, PORTO				
	NEW CALEDONIA	28	1875-3-28	N
EW CALEDONIA: LOYALTY ISLANDS: LIFOU ISLAND				
	AZORES (PORTUGAL)	28	1980-1-1	
AZORES: TERCEIRA, ANGRA DO HEROISMO				
	TONGA	26	1919-4-30	
TONGA ISLANDS				
	ISRAEL	25	1759-10-30	
ISRAEL: ZEFAT (SAFED)				
	IRAQ	24	1864-12-2	
IRAQ: ZURBATIYAH, BADRAH, TURSAQ, BAGHDAD				
	PANAMA	24	1882-9-7	
PANAMA: SAN BLAS ARCHIPELAGO				
	KERMADEC ISLANDS (NEW ZEALAND)	23	1986-10-20	
KERMADEC ISLANDS: RAOUL				
	SOUTH KOREA	22	1643-7-25	
SEA OF JAPAN				
	SLOVENIA	22	1511-3-26	
BALKANS NW: SLOVENIA: IDRIJA,SKOFJA LOKA				
	CANADA	22	1949-8-22	
CANADA: QUEEN CHARLOTTE ISLANDS				
	MOROCCO	21	2004-2-24	M
OROCCO: AL HOCEIMA, IMZOURENE, BENI ABDALLAH				
	HAITI	21	1842-5-7	
HAITI: CAP-HAITIEN				
	NEPAL	21	1505-6-6	
NEPAL-INDIA				
	FIJI	20	1919-1-1	
FIJI ISLANDS				

BANGLADESH	19	1918-7-8		
BANGLADESH: SRIMANGAL				
DOMINICAN REPUBLIC	19	1946-8-4		
DOMINICAN REPUBLIC: NORTHEASTERN COAST				
JAMAICA	19	1899-6-14		
JAMAICA				
BOLIVIA	19	1916-8-25		
BOLIVIA-NORTHERN CHILE				
BULGARIA	18	1904-4-4		
BULGARIA: STRUMA				
ICELAND	17	2000-6-17		
ICELAND: VESTMANNAEYJAR, HELLA				
EGYPT	17	1995-11-22		
EGYPT: NUWAYBI; SAUDI ARABIA; ISRAEL; JORDAN				
AZERBAIJAN	17	1667-11-?		
AZERBAIJAN: SHEMAKHA (SAMAXI)				
CUBA	15	1992-5-25		
CUBA: PILON, MANZANILLO				
KYRGYZSTAN	15	1911-1-3		
KAZAKHSTAN: ALMA-ATA, TURKESTAN; AFGHANISTAN				
GEORGIA	15	1905-10-21		
GEORGIA: CAUCASUS				
SERBIA	15	1980-5-18		
BALKANS NW: SERBIA				
UK	15	1931-6-7		
UK: SCARBOROUGH, GRIMSBY				
HONDURAS	15	1856-8-4		
HONDURAS: COAST				
ROMANIA	15	1986-8-30		
ROMANIA: BUCHAREST, KISHINEV-KAGUL				
SOUTH AFRICA	14	1942-11-10		
SOUTH AFRICA: PRINCE EDWARD ISLAND				
LEBANON	14	1759-11-25	LEBA	
NON-SYRIA: BAALBEC; SYRIA: DAMASCUS, ANTIOCH				
UZBEKISTAN	14	1976-4-8		
UZBEKISTAN: GAZLI				
ARMENIA	14	1988-12-7		
ARMENIA: LENINAKAN, SPITAK, KIROVAKAN				
NORTH MACEDONIA	13	1979-5-24		
BALKANS NW: MACEDONIA: DEBAR				
BRAZIL	13	1986-11-30		
BRAZIL: JOAO CAMARA, NATAL				
TURKMENISTAN	12	1895-7-8		
TURKMENISTAN: UZUN-ADA				
UKRAINE	12	1927-9-11		
UKRAINE: CRIMEA: SEBASTOPOL				
ETHIOPIA	11	1961-6-1		
ETHIOPIA: KARAKORE				
BOSNIA-HERZEGOVINA	11	1990-11-27		
BALKANS NW: BOSNIA-HERZEGOVINA: TITOGRAD				
MONTENEGRO	10	1979-4-15		
BALKANS NW: MONTENEGRO				
CYPRUS	10	1953-9-10		
CYPRUS: PAPHOS				
YEMEN	10	1982-12-13		
YEMEN: DHAMAR				

MARTINIQUE	10	1906-12-3		MARTINIQUE
GERMANY	9	1978-9-3		GERMANY
KAZAKHSTAN	9	1978-3-24		KAZAKHSTAN: ALMA-ATA
TUNISIA	9	1957-2-20	TUNIS	IA: SIDI ABID,SIDI TOUIL (LA MEDJA),CAILLOUX
GUADELOUPE	9	1843-2-8		GUADELOUPE: POINTE-A-PITRE
SAMOA	8	1917-6-26		SAMOA ISLANDS
TANZANIA	8	1910-12-13		TANZANIA: RUKWA
AUSTRIA	8	1998-4-12	AUS	TRIA: ARNOLDSTEIN; SLOVENIA: BOVEC, KOBARID
POLAND	8	2019-7-1		POLAND: KATOWICE
TRINIDAD AND TOBAGO	8	1831-12-3		TRINIDAD & ST. CHRISTOPHER
CONGO	7	2005-12-5		CONGO: KALEMIE
SOUTH GEORGIA AND THE SOUTH SANDWICH ISLANDS	7	1929-6-27		SOUTH SANDWICH ISLANDS
ANTARCTICA	7	1998-3-25		BALLENY ISLANDS
ATLANTIC OCEAN	7	1941-11-25		ATLANTIC OCEAN: NORTHERN
MONGOLIA	7	1905-7-23		MONGOLIA
ERITREA	6	1915-9-23		ERITREA: ASMARA
GHANA	6	1939-6-22		GHANA: ACCRA
NORTH KOREA	6	1668-7-31		NORTH KOREA: YELLOW SEA
VIETNAM	6	1983-6-24		VIETNAM
HUNGARY	5	Unknown		Unknown
JORDAN	5	362-5-24		JORDAN: AL-KARAK
BHUTAN	5	2009-9-21		BHUTAN: TASHIGANG
MICRONESIA, FED. STATES OF	5	1911-8-16		MICRONESIA, FED. STATES OF: CAROLINE ISLANDS
RWANDA	5	2008-2-14		RWANDA: GISENYI
THAILAND	4	1983-4-22		THAILAND: BANGKOK
UGANDA	4	1966-3-20	UGAND	A: KICHWAMBA, BONDIBOGYO; TANZANIA; DR CONGO
MALAWI	4	1989-3-10		MALAWI: SALIMA, DEDZA, MOHINJI
SOUTH SUDAN	4	1990-5-20		SOUTH SUDAN: JUBA, MAYA; UGANDA: NAKURA

NETHERLANDS	4	1992-4-13	THE NETHERLANDS: ROERMOND; GERMANY: BONN, HEINSBERG
ANTIGUA AND BARBUDA	3	1690-4-16	ANTIGUA; SAINT KITTS AND NEVIS
SLOVAKIA	3	Unknown	Unknown
SAUDI ARABIA	3	2009-5-19	SAUDI ARABIA: WESTERN
INDIAN OCEAN	3	1928-3-9	INDIAN OCEAN: S
KENYA	3	1928-1-6	KENYA: SUBUKIA
MALAYSIA	3	1976-7-26	MALAYSIA: SABAH: LAHAD,DATU,KANAK
MOZAMBIQUE	3	2006-2-22	MOZAMBIQUE
LAOS	2	2007-5-16	LAOS: BOKEO; THAILAND: CHIANG RAI, CHIAN SAEN
PACIFIC OCEAN	2	1958-11-4	EAST PACIFIC RIDGE
TOGO	2	Unknown	Unknown
LIBYA	2	1963-2-21	LIBYA: BARCE (AL MARJ)
OMAN	2	Unknown	Unknown
NORWAY	2	1819-8-31	NORWAY: RANA REGION: LUROY
UK TERRITORY	2	1983-11-30	INDIAN OCEAN: CHAGOS ARCHIPELAGO: DIEGO GARCIA
FRENCH GUIANA	2	Unknown	Unknown
SAINT LUCIA	2	Unknown	Unknown
URUGUAY	2	Unknown	Unknown
CANARY ISLANDS	2	Unknown	Unknown
COTE D'IVOIRE	2	Unknown	Unknown
SOLOMON SEA	2	1895-3-6	W. SOLOMON SEA
CAMEROON	2	Unknown	Unknown
DJIBOUTI	2	1989-8-20	DJIBOUTI: GALAFI, YABAKI; ETHIOPIA
SIERRA LEONE	1	Unknown	Unknown
SENEGAL	1	Unknown	Unknown
DOMINICA	1	Unknown	Unknown
IRELAND	1	Unknown	Unknown
SAINT VINCENT AND THE GRENADINES	1	Unknown	Unknown

```
                                    GRENADA                 1      Unknown
Unknown
                                  SRI LANKA                 1      Unknown
Unknown
                                 TASMAN SEA                 1    1892-1-26
TASMAN SEA
                       BRITISH VIRGIN ISLANDS               1      Unknown
Unknown
                                   BARBADOS                 1      Unknown
Unknown
                              FRENCH POLYNESIA              1    1848-7-12
FRENCH POLYNESIA:  TAHITI
                                   ESTONIA                  1      Unknown
Unknown
                                 MONTSERRAT                 1      Unknown
Unknown
                                   KIRIBATI                 1      Unknown
Unknown
                                  GREENLAND                 1      Unknown
Unknown
                                     GUINEA                 1   1983-12-22
GUINEA:  GAOUAL-KOUMBIA
                                      PALAU                 1   1914-10-23
MICRONESIA, FED. STATES OF:  CAROLINE ISLANDS
                     CENTRAL AFRICAN REPUBLIC               1      Unknown
Unknown
                                    BELGIUM                 1      Unknown
Unknown
                                      GABON                 1    1974-9-23
GABON
                                      SUDAN                 1     1993-8-1
SUDAN:  KHARTOUM
          WALLIS AND FUTUNA (FRENCH TERRITORY)              1    1993-3-12
FUTUNA ISLAND
                                  BERING SEA                1    1991-2-21
BERING SEA
                                    BURUNDI                 1      Unknown
Unknown
                               CZECH REPUBLIC               1      Unknown
Unknown
                                  MADAGASCAR                1      Unknown
Unknown
                                     ZAMBIA                 1      Unknown
Unknown
                                    COMOROS                 1      Unknown
Unknown
```

2. Air temperature in Shenzhen 1998-2022

In this problem set, we will examine how air temperature changes in Shenzhen from 1998 to 2022 using the hourly weather data measured at the Baoan International Airport. The data set is from NOAA Integrated Surface Dataset. Download the zipped file Baoan_Weather_1998_2022.zip, unzip it, and move the Baoan_Weather_1998_2022.csv file to your working directory.

Read page 10-11 (POS 88-92 and POS 93-93) of the comprehensive user guide for the detailed format of the air temperature data (use column TMP). Explain how you filter the data in your report.

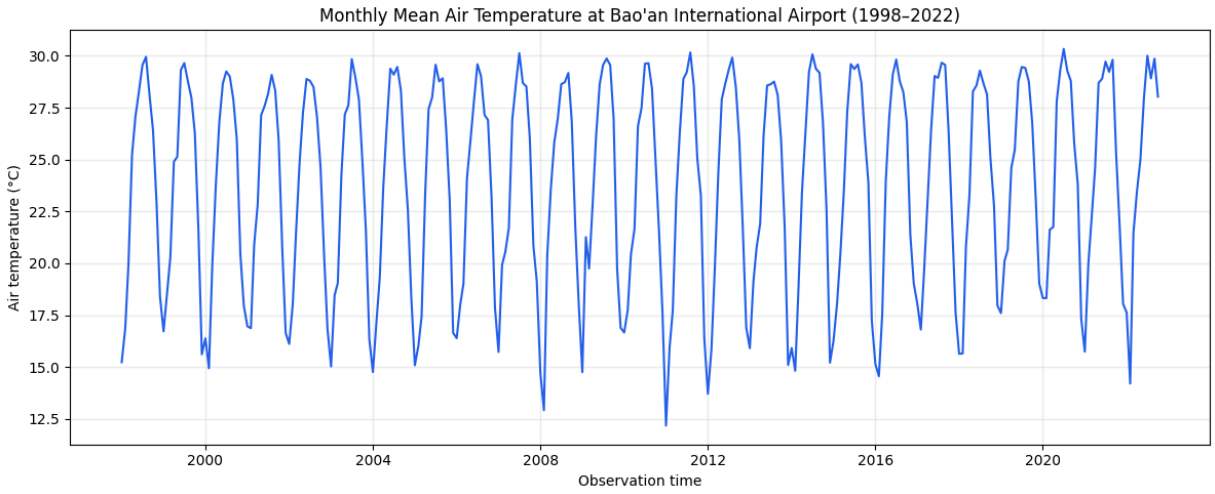
[10 points] Plot monthly averaged air temperature against the observation time. Is there a trend in monthly averaged air temperature from 1998 to 2022?

根据 ISD 指导手册，将 TMP 中的温度值除以 10 转换为摄氏度，剔除 +9999 缺测值，并仅保留质量码为 1 或 5 的观测，因为这两种质量码均表示数据通过全部质量检查

In [ ]:
```
import pandas as pd
weather = pd.read_csv(
    "Baoan_Weather_1998_2022.csv",
    usecols=["DATE", "TMP"],
    parse_dates=["DATE"]
)  # 只读取观测时间和气温，并将 DATE 转为日期时间
tmp_parts = weather["TMP"].str.split(",", n=1, expand=True)  # 拆分气温和质量码
weather["temp_c"] = pd.to_numeric(tmp_parts[0]) / 10  # 将 0.1℃ 转换为 ℃
weather["quality_code"] = tmp_parts[1]  # 提取质量码
weather_clean = weather.loc[
    (tmp_parts[0] != "+9999") & weather["quality_code"].isin(["1", "5"]),
    ["DATE", "temp_c"]
].copy()  # 删除缺测值，仅保留通过全部质量检查的记录
print(weather_clean.head())
print(len(weather_clean))  # 查看筛选后的记录数
```

```
                 DATE  temp_c
0 1998-01-01 00:00:00    18.6
1 1998-01-01 01:00:00    22.0
2 1998-01-01 02:00:00    24.0
3 1998-01-01 03:00:00    22.1
4 1998-01-01 04:00:00    24.0
234864
```

In [62]:
```
import matplotlib.pyplot as plt
monthly_temp = (
    weather_clean.set_index("DATE")["temp_c"]
    .resample("MS")
    .mean()
)  # 按月份分组计算平均气温，日期标记为每月第一天
fig, ax = plt.subplots(figsize=(12, 5))  # 创建画布和坐标轴
ax.plot(
    monthly_temp.index,
    monthly_temp.values,
    color="#2563EB",
    linewidth=1.5
)  # 绘制月平均气温时间序列
ax.set_title("Monthly Mean Air Temperature at Bao'an International Airport (1998–20
ax.set_xlabel("Observation time")  # 设置横轴标题
ax.set_ylabel("Air temperature (°C)")  # 设置纵轴标题
ax.grid(alpha=0.25)  # 添加浅色网格线
fig.tight_layout()  # 自动调整布局，避免文字被截断
plt.show()
```



In [63]:
```python
import numpy as np
import matplotlib.pyplot as plt

monthly_climatology = monthly_temp.groupby(
    monthly_temp.index.month
).transform("mean")  # 计算每个日历月在1998–2022年的平均气温

monthly_anomaly = monthly_temp - monthly_climatology  # 月气温减去对应月份的气候平均
running_anomaly = monthly_anomaly.rolling(12, center=True).mean()  # 计算12个月滑动

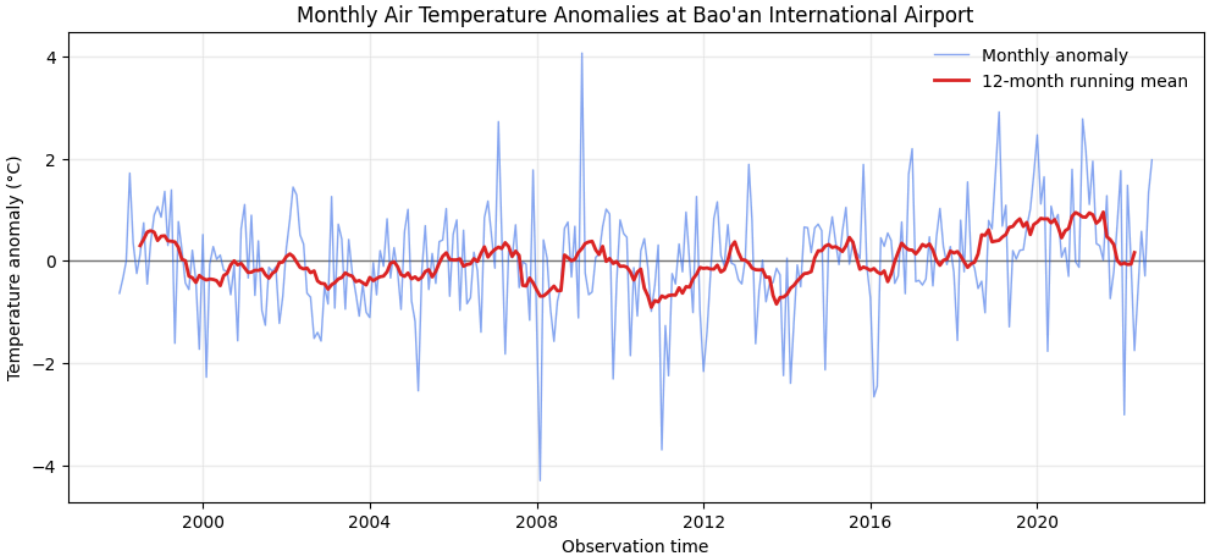
fig, ax = plt.subplots(figsize=(12, 5))

ax.axhline(0, color="#777777", linewidth=1)  # 添加距平零线
ax.plot(
    monthly_anomaly.index,
    monthly_anomaly.values,
    color="#2563EB",
    linewidth=1,
    alpha=0.55,
    label="Monthly anomaly"
)  # 绘制逐月距平

ax.plot(
    running_anomaly.index,
    running_anomaly.values,
    color="#DC2626",
    linewidth=2,
    label="12-month running mean"
)  # 绘制12个月滑动平均，便于观察长期变化

ax.set_title("Monthly Air Temperature Anomalies at Bao'an International Airport")
ax.set_xlabel("Observation time")
ax.set_ylabel("Temperature anomaly (°C)")
ax.grid(alpha=0.2)
ax.legend(frameon=False)
```

Out[63]: <matplotlib.legend.Legend at 0x20553d3fe00>



图中可以看到非常明显的季节变化：夏季气温高、冬季气温低。 去除季节循环后，月平均气温距平仍然表现出明显的年际波动。12 个月滑动平均显示，研究期内冷暖阶段交替出现，并非持续升温；但后期正距平相对较多，线性拟合表明 1998—2022 年整体呈弱升温趋势，升温速率约为 0.023℃/年。

3. Global collection of hurricanes

The International Best Track Archive for Climate Stewardship (IBTrACS) project is the most complete global collection of tropical cyclones available. It merges recent and historical tropical cyclone data from multiple agencies to create a unified, publicly available, best-track dataset that improves inter-agency comparisons. IBTrACS was developed collaboratively with all the World Meteorological Organization (WMO) Regional Specialized Meteorological Centers, as well as other organizations and individuals from around the world.

In this problem set, we will use all storms available in the IBTrACS record since 1842. Download the zipped file ibtracs.ALL.list.v04r00.zip, unzip it, and move the ibtracs.ALL.list.v04r00.csv file to your working directory. Read Column Variable Descriptions for variables in the file. Examine the first few lines of the file.

Below we provide an example to load the file as a pandas dataframe. Think about the options being used and why, and modify when necessary.

df = pd.read_csv('ibtracs.ALL.list.v04r00.csv', usecols=range(17), skiprows=[1, 2], parse_dates=['ISO_TIME'], na_values=['NOT_NAMED', 'NAME']) df.head()

3.1 [5 points] Group the data on Storm Identifie (SID), report names (NAME) of the 10 largest hurricanes according to wind speed (WMO_WIND).

In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
```

```
df = pd.read_csv(
    "ibtracs.ALL.list.v04r00.csv",
    usecols=range(17),  # 只读取前17列
    skiprows=[1],  # 跳过第二行的单位说明
    skipinitialspace=True,  # 去除分隔符后的空格，正确识别缺测值
    parse_dates=["ISO_TIME"],  # 将观测时间转为日期时间
    na_values=["NOT_NAMED", "NAME"],  # 将无名风暴标记为缺失值
    low_memory=False
)

for col in ["LAT", "LON", "WMO_WIND"]:
    df[col] = pd.to_numeric(df[col], errors="coerce")  # 转为数值，无效值设为NaN

storm_max = df.groupby("SID").agg(
    NAME=("NAME", "first"),
    MAX_WIND=("WMO_WIND", "max")
)  # 按风暴ID提取名称和最大风速

top10_names = storm_max.nlargest(10, "MAX_WIND")["NAME"].fillna("NOT_NAMED")  # 取前
print(top10_names.to_string(index=False, header=False))  # 只输出风暴名称
```

```
 PATRICIA
    ALLEN
NOT_NAMED
  GILBERT
    LINDA
    WILMA
   DORIAN
    MITCH
     RITA
     RICK
```

3.2 [5 points] List the wind speed (WMO_WIND) of the 20 strongest-wind hurricanes.

In [2]:
```
top20_winds = storm_max["MAX_WIND"].nlargest(20)  # 每个风暴取最大风速，再选前20个
print(top20_winds.to_string())  # 输出SID及对应最大风速（节）
```

```
SID
2015293N13266    185.0
1980214N11330    165.0
1935241N23291    160.0
1988253N12306    160.0
1997253N12255    160.0
2005289N18282    160.0
2019236N10314    160.0
1998295N12284    155.0
2005261N21290    155.0
2009288N07267    155.0
2017242N16333    155.0
1932304N15305    150.0
1955265N13307    150.0
1959249N19181    150.0
1969226N18280    150.0
1977242N27272    150.0
1979238N12324    150.0
1992230N11325    150.0
1994222N11267    150.0
2005236N23285    150.0
```

3.3 [5 points] Count all datapoints by Basin.

In [3]:
```
basin_counts = df["BASIN"].value_counts(dropna=False)  # 统计各海域记录数，包括海域
print(basin_counts.to_string())  # 输出各海域的数据点数量
```

```
BASIN
WP     236576
SI     160668
NaN    124881
SP      67119
EP      62412
NI      55402
SA        119
```

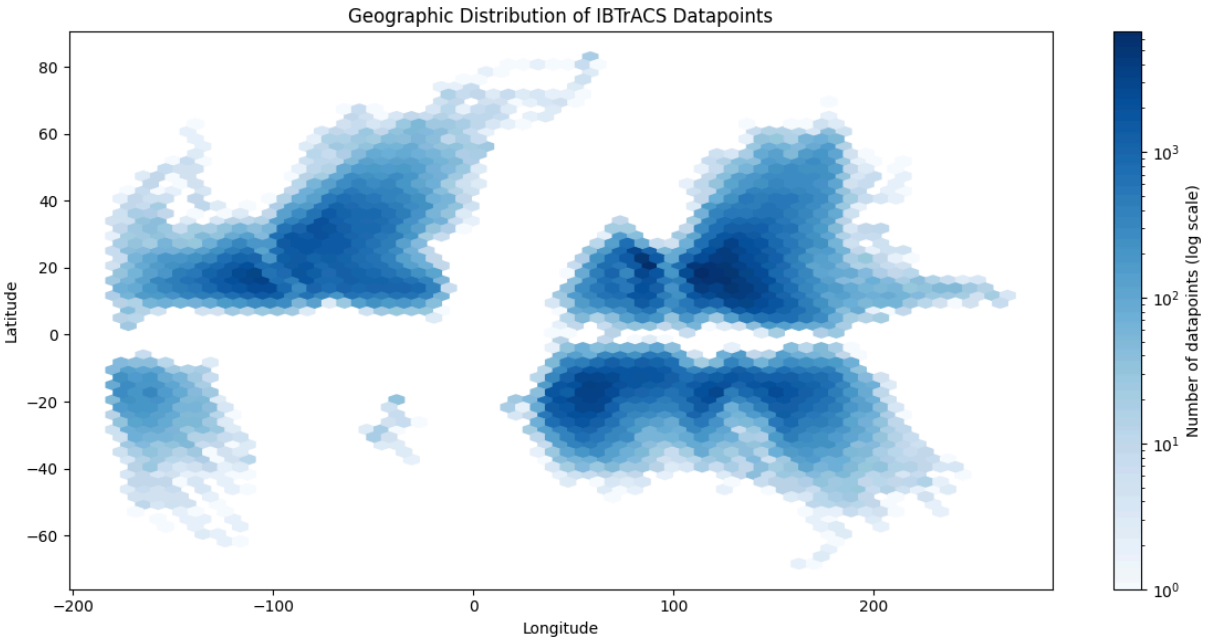
3.4 [5 points] Make a hexbin plot of the location of datapoints in Latitude and Longitude.

In [4]:
```
fig, ax = plt.subplots(figsize=(12, 6))

hb = ax.hexbin(
    df["LON"],
    df["LAT"],
    gridsize=60,
    bins="log",
    mincnt=1,
    cmap="Blues"
)  # 用六边形统计经纬度位置的数据点密度

fig.colorbar(hb, ax=ax, label="Number of datapoints (log scale)")  # 添加对数密度颜
ax.set_title("Geographic Distribution of IBTrACS Datapoints")
ax.set_xlabel("Longitude")
ax.set_ylabel("Latitude")
```

```
fig.tight_layout()
plt.show()
```



3.5 [5 points] Find Typhoon Mangkhut (from 2018) and plot its track.

In [5]:
```
import plotly.graph_objects as go

mangkhut = df[
    df["NAME"].eq("MANGKHUT") & df["SEASON"].eq(2018)
].sort_values("ISO_TIME")  # 筛选2018年台风山竹并按时间排序

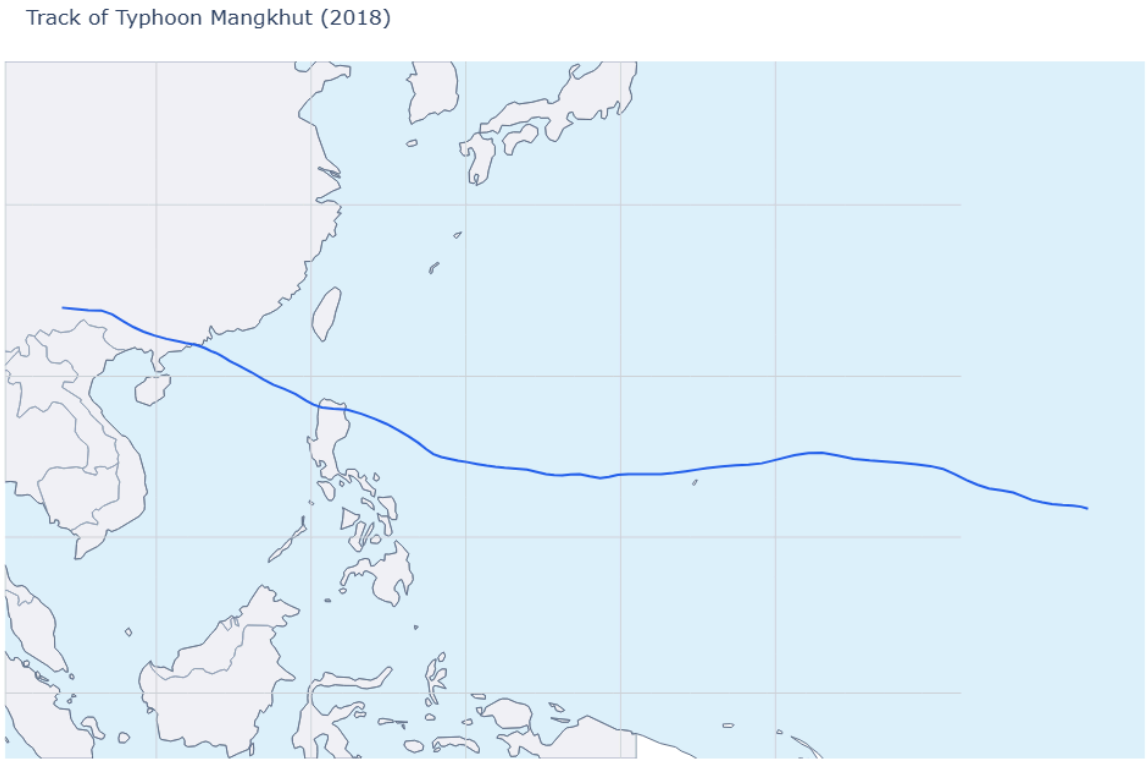
fig = go.Figure(go.Scattergeo(
    lon=mangkhut["LON"],
    lat=mangkhut["LAT"],
    mode="lines",
    line=dict(color="#2563EB", width=2)
))  # 在地图上绘制台风路径

fig.update_geos(
    scope="asia",
    fitbounds="locations",  # 根据轨迹自动调整地图范围
    showcoastlines=True,
    coastlinecolor="#64748B",
    showland=True,
    landcolor="#F3F4F6",
    showocean=True,
    oceancolor="#E0F2FE",
    showcountries=True,
    countrycolor="#94A3B8",
    lataxis=dict(showgrid=True, gridcolor="#D1D5DB", dtick=10),
    lonaxis=dict(showgrid=True, gridcolor="#D1D5DB", dtick=10)
)  # 添加海岸线、国界、陆地和海洋

fig.update_layout(
    title="Track of Typhoon Mangkhut (2018)",
    width=1000,
```

```
    height=650,
    margin=dict(l=20, r=20, t=60, b=20),
    showlegend=False
)

fig.show(renderer="png")  # 显示普通静态地图
```



3.6 [5 points] Create a filtered dataframe that contains only data since 1970 from the Western North Pacific (“WP”) and Eastern North Pacific (“EP”) Basin. Use this for the rest of the problem set.

In [6]:
```
df_filtered = df[
    (df["ISO_TIME"].dt.year >= 1970) & df["BASIN"].isin(["WP", "EP"])
].copy()  # 筛选1970年以来WP和EP海域的记录

print("Filtered datapoints:", len(df_filtered))  # 输出筛选后的记录数
print("Basins:", df_filtered["BASIN"].unique())  # 核对海域
print("Date range:", df_filtered["ISO_TIME"].min(), "to", df_filtered["ISO_TIME"].m
```

```
Filtered datapoints: 176352
Basins: <StringArray>
['WP', 'EP']
Length: 2, dtype: str
Date range: 1970-02-19 00:00:00 to 2022-10-12 18:00:00
```

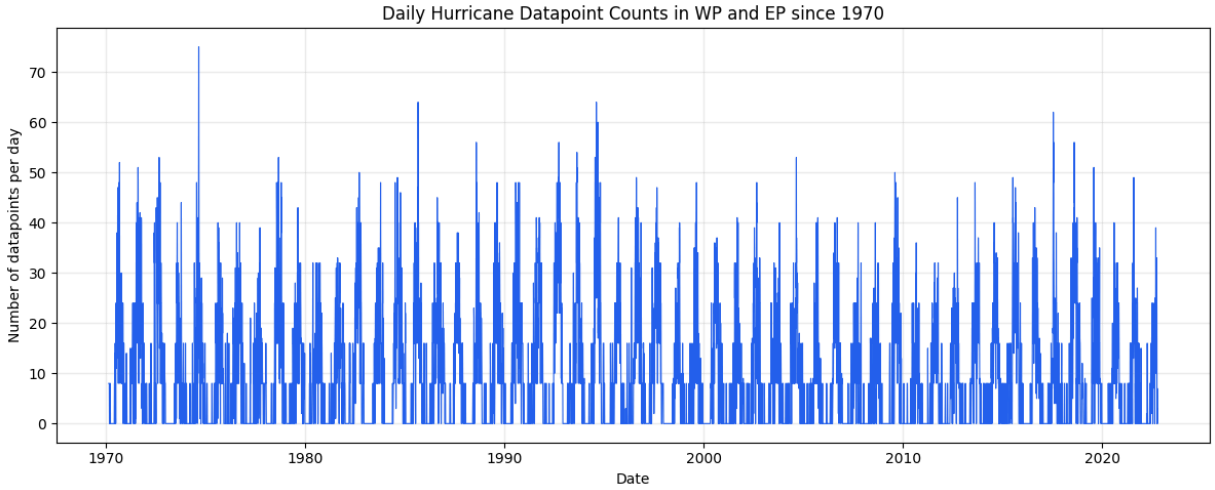
3.7 [5 points] Plot the number of datapoints per day.

In [7]:
```
daily_counts = df_filtered.set_index("ISO_TIME").resample("D").size()  # 统计每天的

fig, ax = plt.subplots(figsize=(12, 5))
ax.plot(daily_counts.index, daily_counts.values, color="#2563EB", linewidth=0.8)  #
```

```
ax.set_title("Daily Hurricane Datapoint Counts in WP and EP since 1970")
ax.set_xlabel("Date")
ax.set_ylabel("Number of datapoints per day")
ax.grid(alpha=0.25)

fig.tight_layout()
plt.show()
```



3.8 [5 points] Calculate the climatology of datapoint counts as a function of day of year. The day of year is the sequential day number starting with day 1 on January 1st.

In [8]:
```
daily_climatology = daily_counts.groupby(daily_counts.index.dayofyear).mean()  # 计
daily_climatology.index.name = "Day of year"  # 设置索引名称

print(daily_climatology.head(10).to_string())  # 查看前10个年内日的气候态
```

```
Day of year
1     1.596154
2     1.384615
3     1.423077
4     1.788462
5     2.019231
6     2.326923
7     2.173077
8     2.788462
9     2.653846
10    2.634615
```

3.9 [5 points] Calculate the anomaly of daily counts from the climatology.

In [9]:
```
daily_anomaly = daily_counts - daily_climatology.loc[
    daily_counts.index.dayofyear
].to_numpy()  # 每日计数减去对应年内日的气候平均计数

print(daily_anomaly.head(10).to_string())  # 查看前10天的距平
```

```
ISO_TIME
1970-02-19    7.018868
1970-02-20    7.094340
1970-02-21    7.188679
1970-02-22    7.377358
1970-02-23    7.490566
1970-02-24    7.547170
1970-02-25    7.396226
1970-02-26    7.245283
1970-02-27    6.905660
1970-02-28   -0.150943
Freq: D
```

3.10 [5 points] Resample the anomaly timeseries at annual resolution and plot. So which years stand out as having anomalous hurricane activity?

In [10]:
```python
annual_anomaly = daily_anomaly.resample("YE").mean()  # 将每日距平重采样为年平均距平

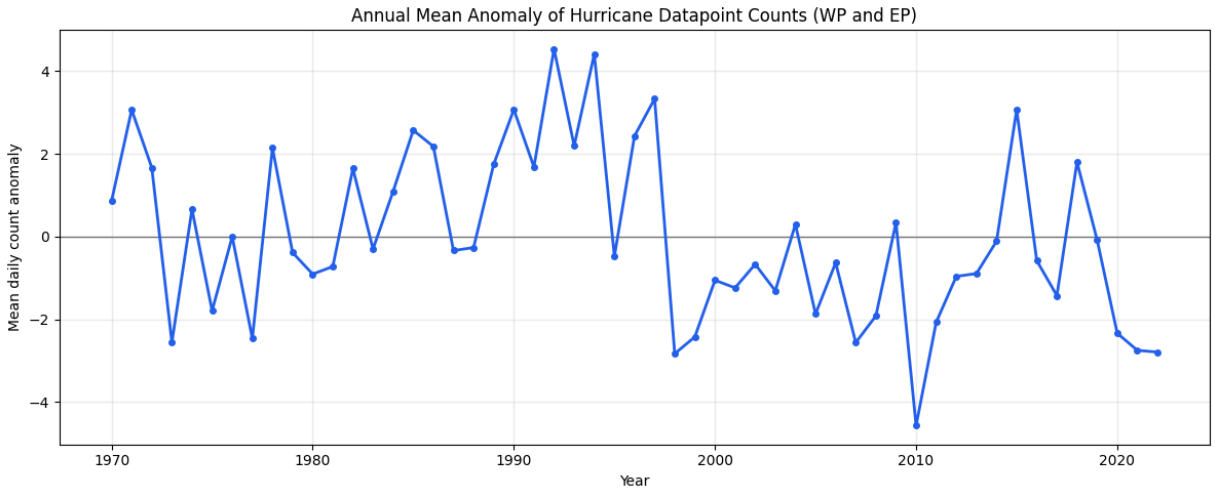
standout_years = annual_anomaly.abs().nlargest(5)  # 找出绝对距平最大的5年
print(annual_anomaly.loc[standout_years.index].to_string())  # 输出异常最突出的年份

fig, ax = plt.subplots(figsize=(12, 5))
ax.axhline(0, color="#777777", linewidth=1)  # 添加零距平参考线
ax.plot(
    annual_anomaly.index.year,
    annual_anomaly.values,
    color="#2563EB",
    linewidth=2,
    marker="o",
    markersize=4
)  # 绘制年平均距平

ax.set_title("Annual Mean Anomaly of Hurricane Datapoint Counts (WP and EP)")
ax.set_xlabel("Year")
ax.set_ylabel("Mean daily count anomaly")
ax.grid(alpha=0.25)

fig.tight_layout()
plt.show()
```

```
ISO_TIME
2010-12-31   -4.570616
1992-12-31    4.537501
1994-12-31    4.415686
1997-12-31    3.330754
1990-12-31    3.073220
```



1992 年和 1994 年的正距平最为突出，表明白天风数据点计数所反映的气旋活动整体偏强；2010 年的负距平最大，表明气旋活动整体偏弱。此外，1997、1990 和 2015 年也表现为较明显的正距平。

4. Explore a data set

4.1 [5 points] Load the csv, XLS, or XLSX file, and clean possible data points with missing values or bad quality.

In [11]:
```
import pandas as pd

n2o_data = pd.read_csv("adr_n2o_clean.csv", parse_dates=["time"])  # 直接读取CSV
original_count = len(n2o_data)  # 保存清洗前的记录数

n2o_data["n2o_ppb"] = pd.to_numeric(n2o_data["n2o_ppb"], errors="coerce")  # 转为数
n2o_data = n2o_data.dropna(subset=["time", "n2o_ppb"])  # 删除缺失记录
n2o_data = n2o_data[n2o_data["n2o_ppb"] > 0]  # 删除非正浓度
n2o_data = n2o_data.drop_duplicates(subset=["time"], keep="first")  # 删除重复时间
n2o_data = n2o_data.sort_values("time").reset_index(drop=True)  # 按时间排序

print("Removed records:", original_count - len(n2o_data))  # 输出删除的记录数
print("Valid records:", len(n2o_data))  # 输出有效记录数
print(n2o_data.head().to_string(index=False))  # 查看前5行
```

```
Removed records: 0
Valid records: 6093
                time   n2o_ppb
1978-07-01 01:05:00 299.72840
1978-07-01 02:06:00 304.85797
1978-07-01 03:07:00 304.85797
1978-07-01 07:10:00 301.84058
1978-07-01 08:11:00 304.85797
```

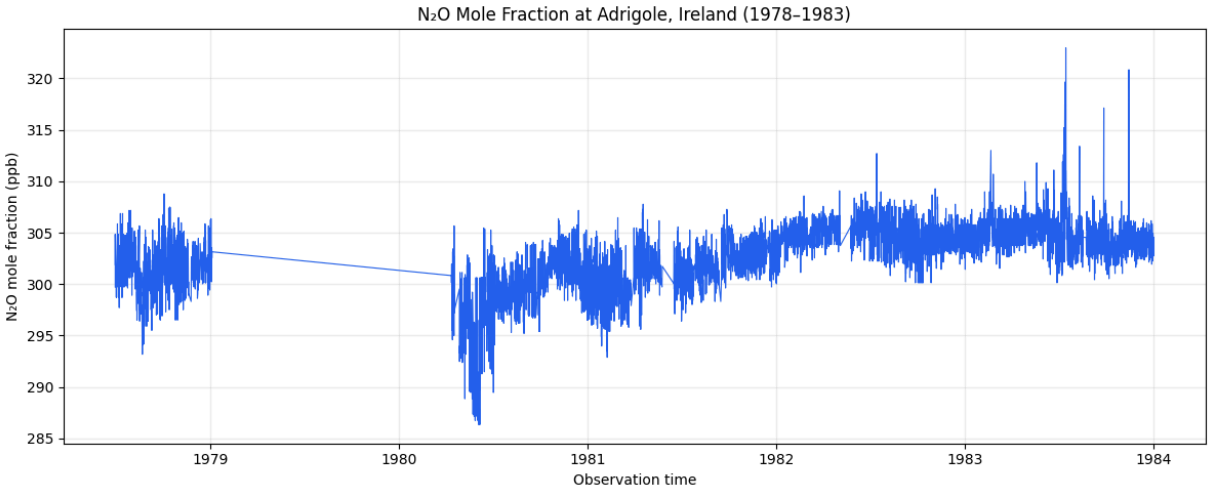
4.2 [5 points] Plot the time series of a certain variable.

In [12]:
```
import matplotlib.pyplot as plt

n2o_plot = pd.read_csv("adr_n2o_clean.csv", parse_dates=["time"])  # 读取清洗后的CS
```

```
fig, ax = plt.subplots(figsize=(12, 5))
ax.plot(n2o_plot["time"], n2o_plot["n2o_ppb"], color="#2563EB", linewidth=0.8)  # 绘
ax.set_title("N₂O Mole Fraction at Adrigole, Ireland (1978–1983)")
ax.set_xlabel("Observation time")
ax.set_ylabel("N₂O mole fraction (ppb)")
ax.grid(alpha=0.25)

fig.tight_layout()
plt.show()
```



4.3 [5 points] Conduct at least 5 simple statistical checks with the variable, and report your findings.

In [13]:
```
import numpy as np

n2o_stats_data = pd.read_csv("adr_n2o_clean.csv", parse_dates=["time"])  # 读取清洗
values = n2o_stats_data["n2o_ppb"]  # 提取N₂O浓度列

x_years = (n2o_stats_data["time"] - n2o_stats_data["time"].min()).dt.days / 365.25
trend = np.polyfit(x_years, values, 1)[0]  # 计算线性变化趋势

statistics = pd.Series({
    "Valid observations": values.count(),
    "Mean (ppb)": values.mean(),
    "Median (ppb)": values.median(),
    "Standard deviation (ppb)": values.std(),
    "Minimum (ppb)": values.min(),
    "25th percentile (ppb)": values.quantile(0.25),
    "75th percentile (ppb)": values.quantile(0.75),
    "Maximum (ppb)": values.max(),
    "Skewness": values.skew(),
    "Linear trend (ppb/year)": trend
})  # 计算描述性统计和变化趋势

print(statistics.round(4).to_string())  # 输出统计结果
print("Time range:", n2o_stats_data["time"].min(), "to", n2o_stats_data["time"].max
```

```
Valid observations          6093.0000
Mean (ppb)                   302.6080
Median (ppb)                 302.9470
Standard deviation (ppb)       3.0711
Minimum (ppb)                286.3513
25th percentile (ppb)        300.9354
75th percentile (ppb)        304.6568
Maximum (ppb)                322.9624
Skewness                      -0.8176
Linear trend (ppb/year)        0.8217
Time range: 1978-07-01 01:05:00 to 1983-12-31 20:05:00
```

清洗后保留 **6093 条**有效观测，原始数据没有缺失值、非正浓度或重复时间，因此删除 **0 条**记录。N_2O 平均浓度为 **302.61 ppb**，中位数为 **302.95 ppb**，标准差约为 **3.07 ppb**，观测范围为 **286.35—322.96 ppb。**偏度约为 **−0.82**，说明分布呈一定左偏。简单线性拟合得到约 **0.822 ppb/年**的上升趋势；该结果仅描述 1978—1983 年样本期内的总体变化，不能仅凭此判断趋势具有统计显著性。